Module Diffusion

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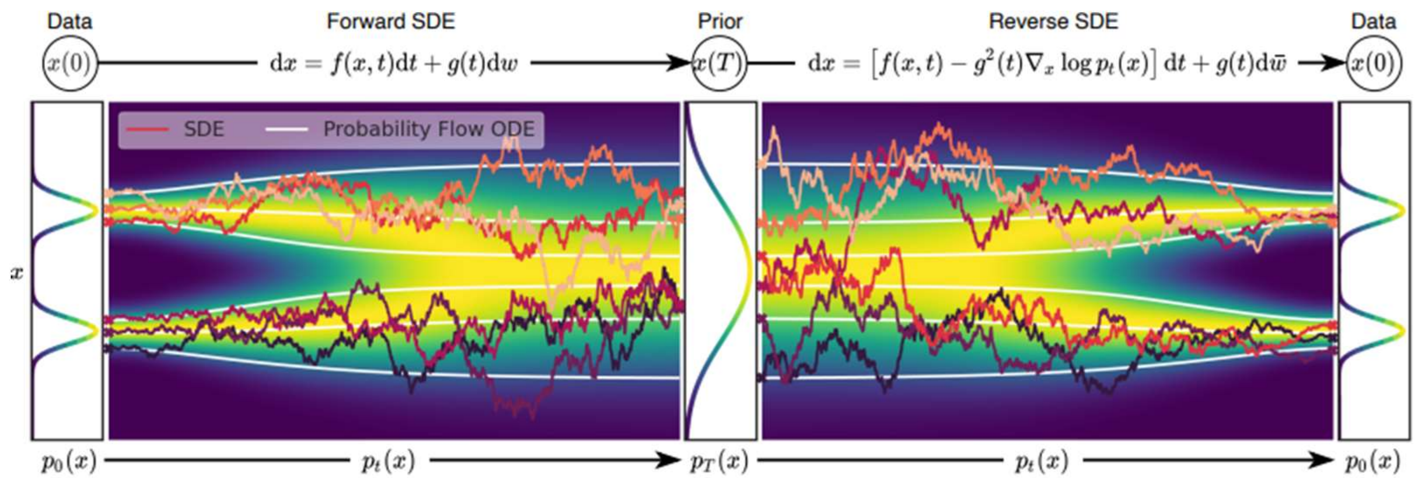
Introduction to Diffusion

- Diffusion models are a class of models that use the concept of diffusion noise
- Stable diffusion is a denoising diffusion model
- Stable diffusion is one type of diffusion model architecture

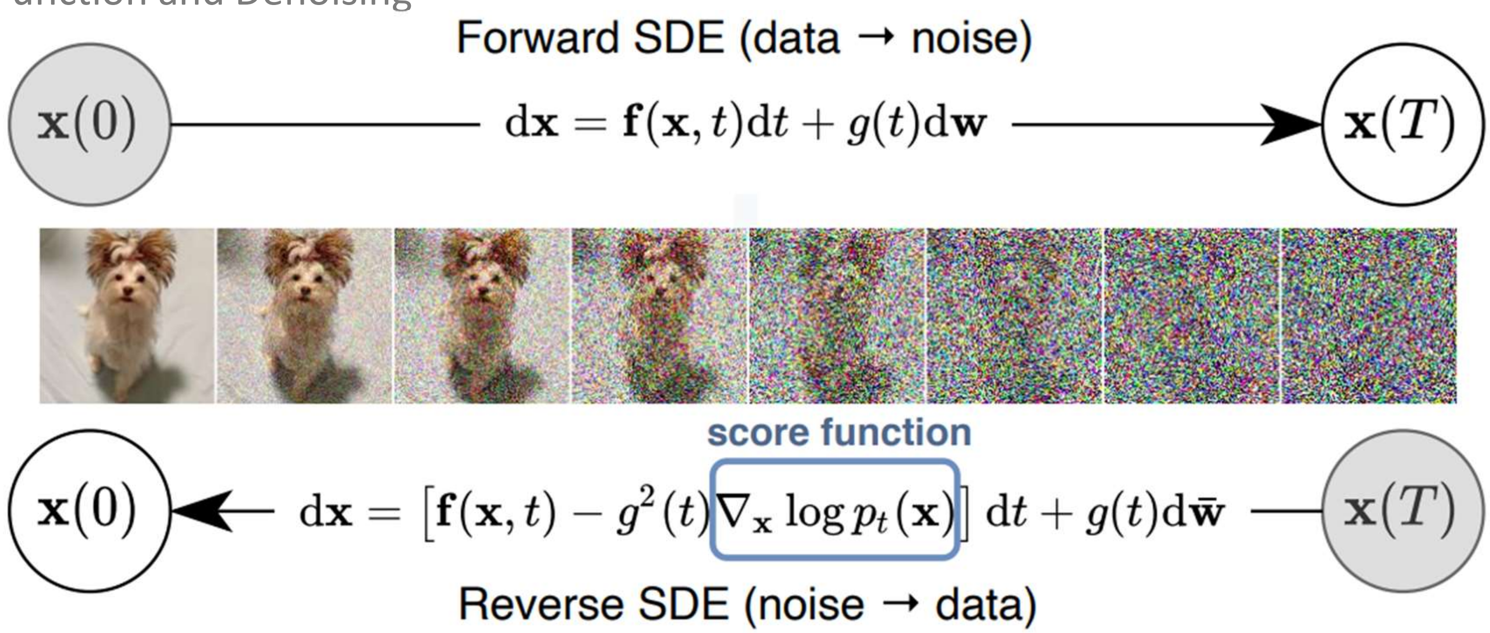
How Diffusion Models Work

1. Diffusion models generate data by reversing a noise process.
2. Inspired by Stochastic Differential Equations (SDEs).



How Diffusion Models Work

1. Forward SDE (Data to Noise)
2. Reverse SDE (Noise to Data)
3. Score Function and Denoising

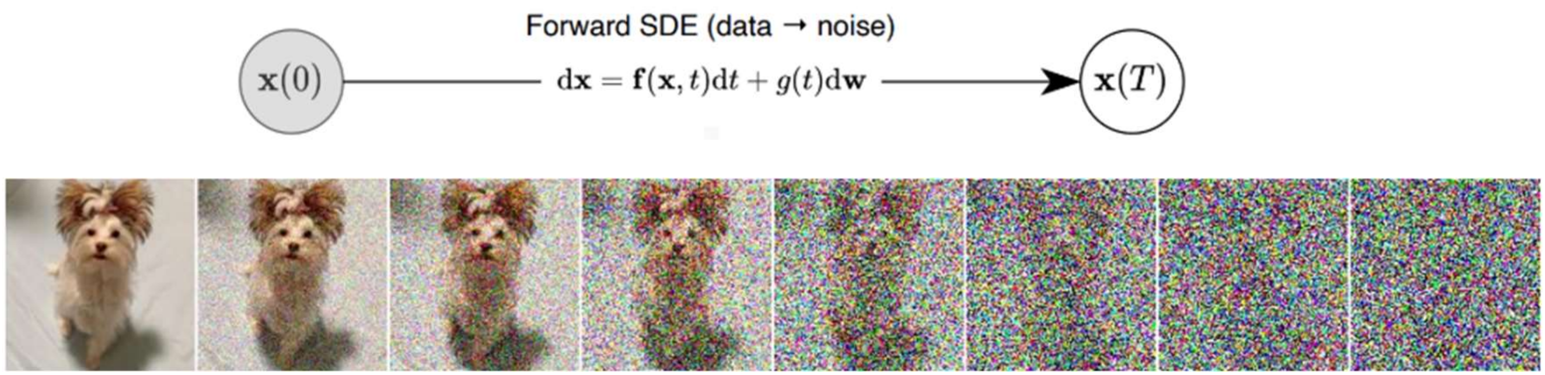


Forward SDE

- The process gradually adds noise to data over time.
- Governed by the reverse equation:

$$dx = f(x,t)dt + g(t)d\mathbf{w}$$

- W term represents a Wiener process (random noise) (aka Brownian)

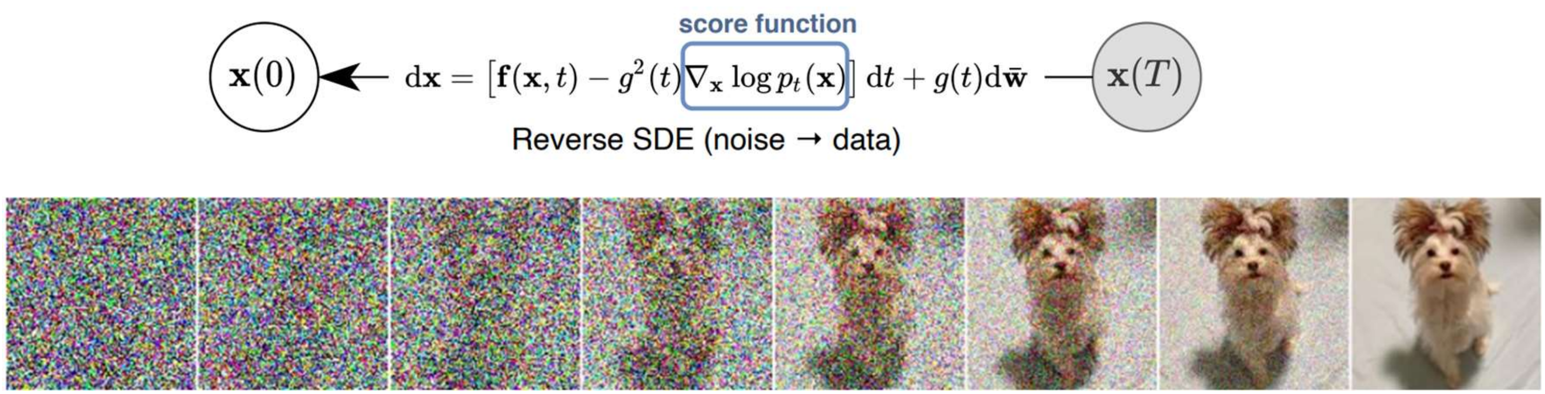


Score Function and Denoising

- The score function of a distribution is given by $f(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \log p_{\text{data}}(\boldsymbol{x})$
- The score function estimates the gradient of log-probability density.
- Helps guide noisy samples back to the data distribution.
- Trained using score matching techniques.

Reverse SDE (Noise to Data)

- The goal is to reverse the noisy transformation to recover data.
- Governed by the reverse equation:
  $dx = [f(x,t) - g^2(t)\nabla_x \log p_t(x)]dt + g(t)d\bar{\mathbf{w}}$
- Requires the score function which guides the process
  $\nabla_x \log p_t(x)$



Overview of Stable Diffusion

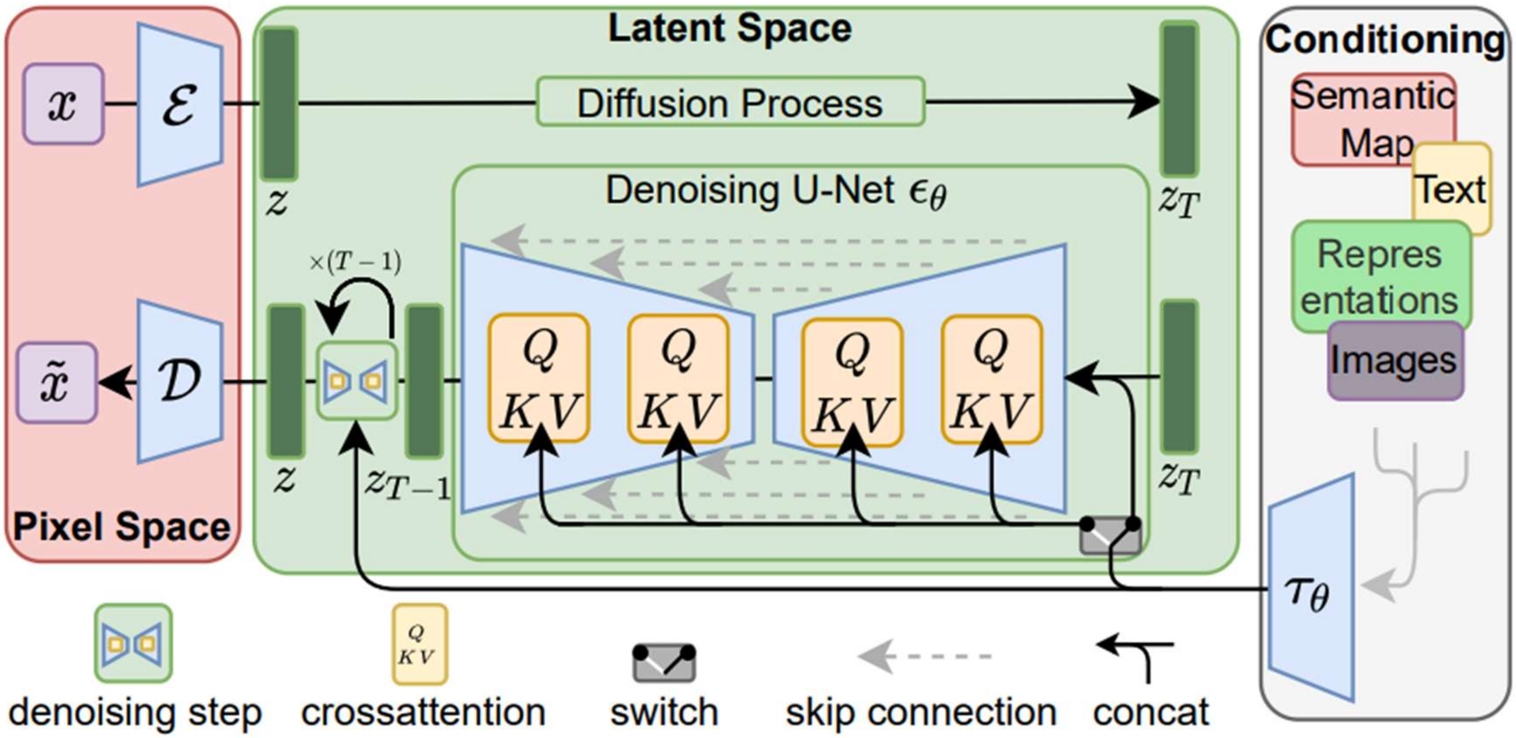
Purpose

- Stable Diffusion is **primarily** used for **generating** images from text **prompts** (**text-to-image**) but can **also** perform other **image transformation** tasks such as **inpainting**, **style transfer**, and **super-resolution.**

Structure

- Stable Diffusion **operates** as a **diffusion model** that iteratively **refines** an **image** starting from **random noise.** The **process** is **guided** by the **input text prompt**, ultimately **forming** a coherent and **high-quality image.**

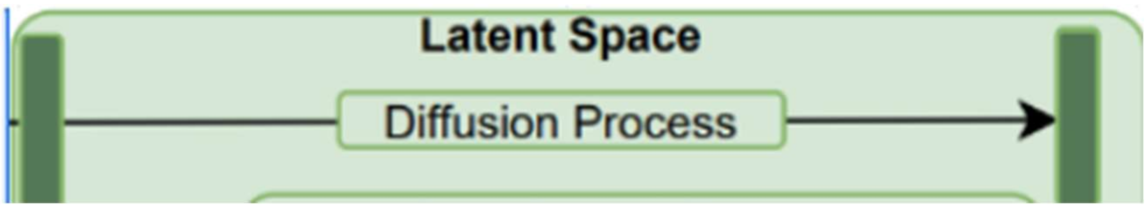
Overview of Stable Diffusion



Stable Diffusion

- Stable Diffusion is a **latent diffusion model.**

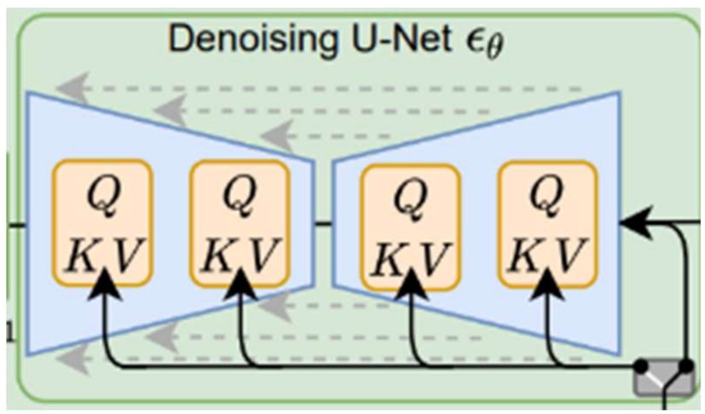
- Instead of directly denoising pixels, it **compresses images** into a **lower-dimensional latent space** using a Variational Autoencoder (VAE).



Stable Diffusion

- It does **not explicitly solve an SDE** but instead uses a **U-Net** to learn the noise removal process in a **discrete manner.**

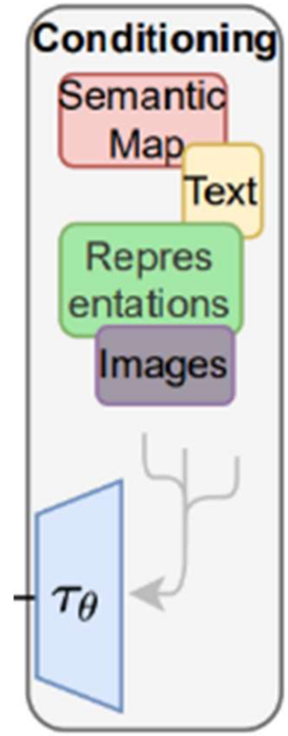
- The **U-Net learns** to **approximate** the **score function** $\nabla_x \log p_t(x)$



Text Encoder

- Stable Diffusion integrates a text encoder, typically based on **CLIP** (Contrastive Language-Image Pretraining), to **interpret** and **embed** text prompts.

- CLIP creates a **shared semantic space** for **textual** and **visual concepts**, enabling the **model** to **translate** textual **descriptions** into **meaningful** visual **features**.



Stable Diffusion Summary

- **Stable Diffusion** uses a denoising diffusion process rather than GAN's adversarial approach, starting with random noise that gradually transforms into a coherent image.
- It employs a **U-Net neural network** to predict the optimal denoising path at each step, guided by the text prompt's requirements.
- The model creates a **probabilistic mapping** between **noise** and **images**, learning how to systematically r**emove noise** in a way that aligns with the **semantic content specified in the text prompt.**